Name

Mathematics Extension 1 Year 11

Further Inequalities

Book 1

Version: 261028

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

ME1-11-01 solves problems involving inequalities, functions and their inverses, graphical relationships between functions, and parametric equations

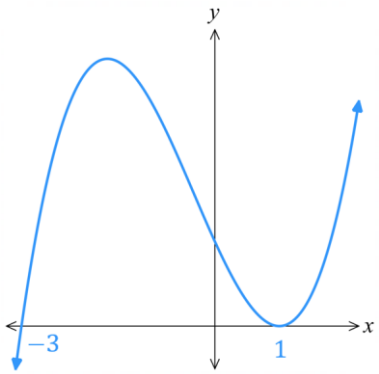
Inequalities

- Solve cubic inequalities where the cubic is expressed as a product of linear factors
- Solve inequalities involving rational expressions with variables in the denominator
- Solve absolute value inequalities of the form $|ax + b| \geq k$, $|ax + b| \leq k$, $|ax + b| < k$, $|ax + b| > k$, where a, b and k are constants, using algebraic and graphical methods or the characterisation of $|x|$ as the distance of x from the origin

Cubic Inequalities

Identify domain where $y > 0$ or $y < 0$

a



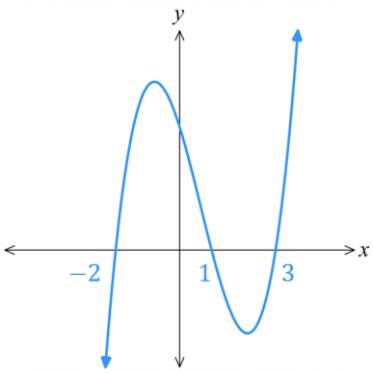
$y > 0$:

$-3 < x < 1$

$y < 0$:

$x < -3, \ x > 1$

b



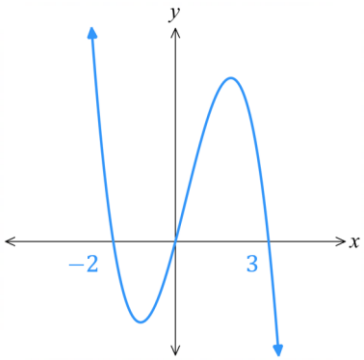
$y > 0$:

$-2 < x < 1, \ x > 3$

$y < 0$:

$x < -2, \ 1 < x < 3$

c



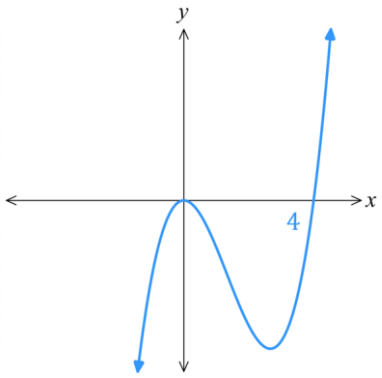
$y > 0$:

$x < -2, \ 0 < x < 3$

$y < 0$:

$-2 < x < 0, \ x > 3$

d



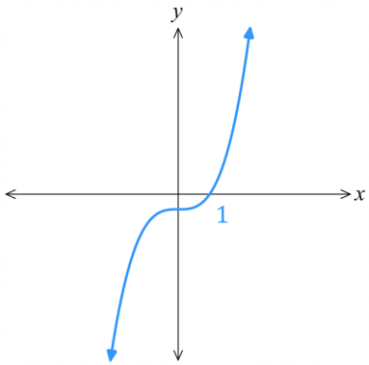
$y > 0$:

$x > 4$

$y < 0$:

$x < 0, \ 0 < x < 4$

e



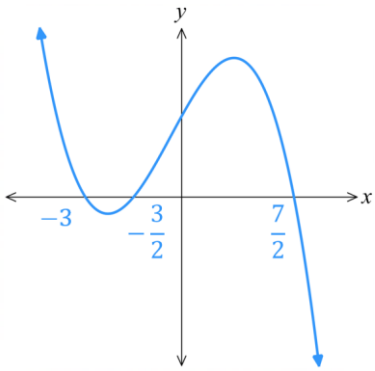
$y > 0$:

$x > 1$

$y < 0$:

$x < 1$

f



$y > 0$:

$x < -3, \ -\frac{3}{2} < x < \frac{7}{2}$

$y < 0$:

$-3 < x < -\frac{3}{2}, \ x > \frac{7}{2}$

Cubic Inequalities

We solve cubic inequalities the same way as we solve quadratic inequalities:

1. Sketch the cubic curve $f(x)$, **showing all x-intercepts**.
2. Examine where the graph is above or below the x axis.

Example Solve cubic inequalities given in factored form

Solve $(x+2)(x-1)(x-3) \geq 0$

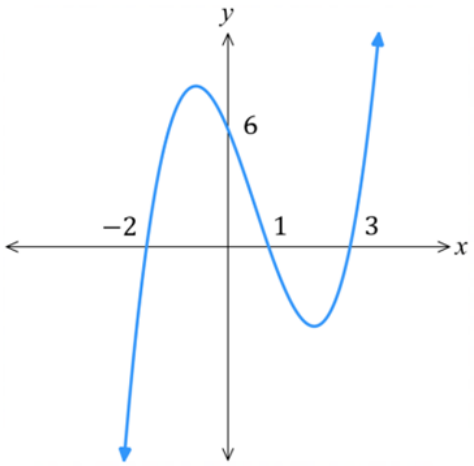
1.

Leading term is x^3,

$x \to \infty, y \to \infty$

x-int:

$x = 2, -1, -3$



2.

$-2 < x < 1, \ x > 3$

Guided Practice

Solve:

a $(x-1)(x+2)(x-3) \leq 0$

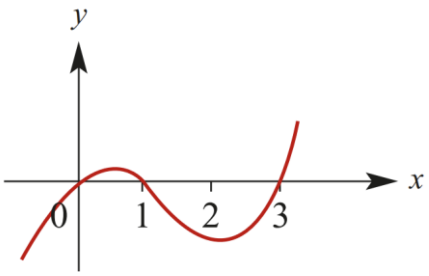
$1 \leq x \leq 3, \ x \leq -2$

b $-x(x+2)(x-3) > 0$

$0 < x < 3, \ x < -2$

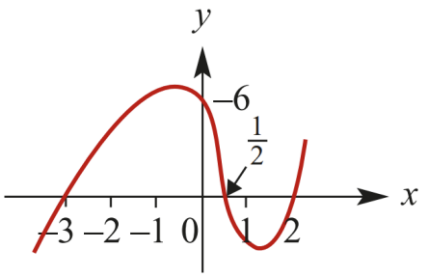
1 Solve these cubic inequalities by referring to the given graph.

a $x(x-1)(x-3) \geq 0$



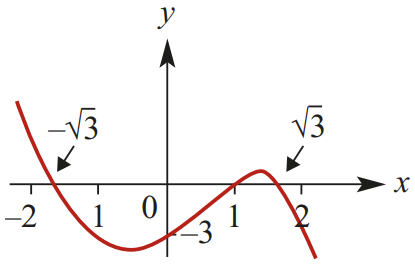
$0 \leq x \leq 1, \quad x \geq 3$

b $(2x-1)(x-2)(x+3) < 0$



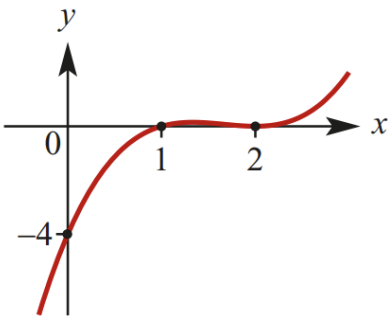
$x < -3, \quad 0.5 < x < 2$

c $(x-1)(3-x^2) \geq 0$



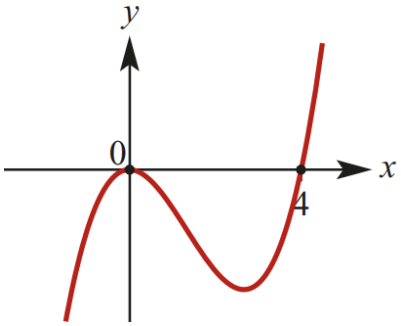
$x \leq -\sqrt{3}, \quad 1 \leq x \leq \sqrt{3}$

d $(x-1)(x-2)^2 \geq 0$



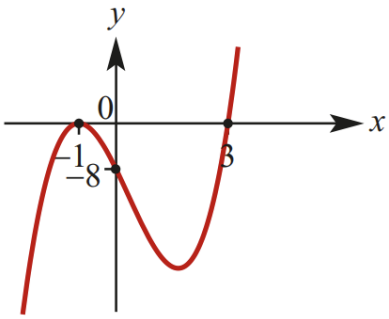
$x \geq 1$

e $x^2(x-4) < 0$



$x \leq 4, \quad x \neq 0$

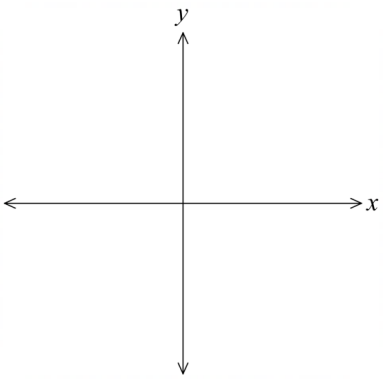
f $2(x+1)^2(x-3) < 0$



$x < 3, \quad x \neq -1$

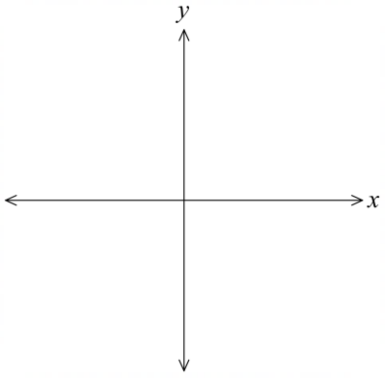
2 Solve these cubic inequalities.

a $(x-1)(x+1)(x-4) \leq 0$



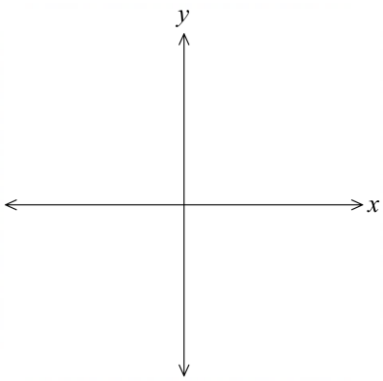
$1 \leq x \leq 4, \quad x \leq -1$

b $(x+2)(x-3)(x+6) > 0$



$-6 < x < -2, \quad x > 3$

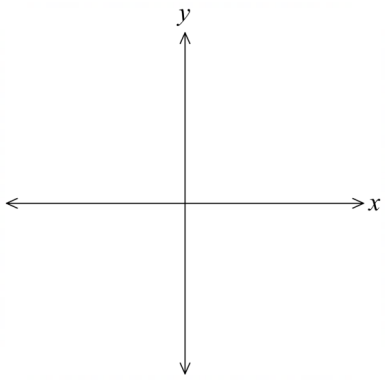
c $-(x-1)(x-2)(x-3) < 0$



$1 < x < 2, \quad x > 3$

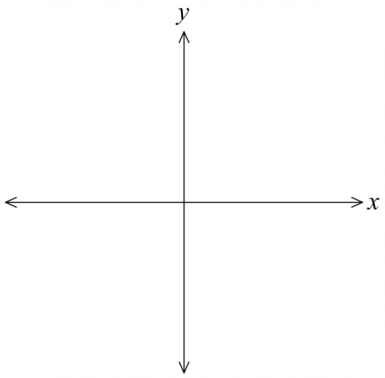
3 Solve these cubic inequalities. Start by moving all terms to the LHS.

a $x^3 > 4x$



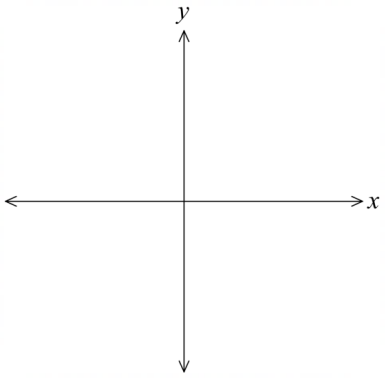
$-2 < x < 0,\ x > 2$

b $x^3 < 5x^2$



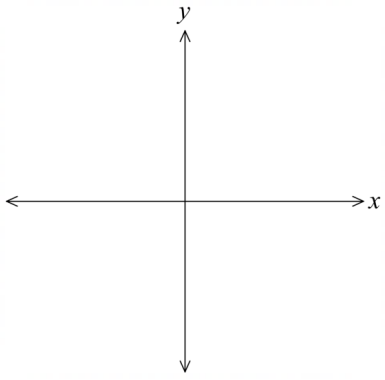
$x < 5,\ x \neq 0$

c $x^3 + 4x \leq 4x^2$



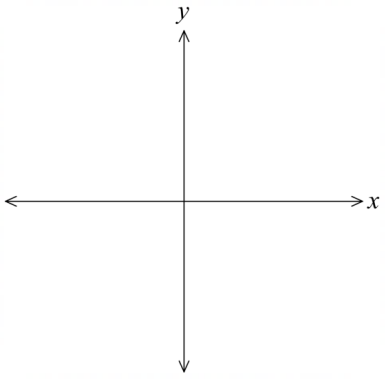
$x \leq 0,\ x = 2$

d $x^3 - 6x^2 + x \geq 6$



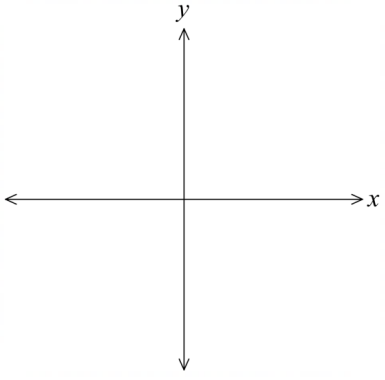
$x \geq 6$

e $2x^3 - 6x^2 - 4x < -12$



$x < -\sqrt{2},\ \sqrt{2} < x < 3$

f $x^4 - 13x^2 + 36 < 0$



$-3 < x < -2,\ 2 < x < 3$

Rational Inequalities

ⓘ Rational Inequalities

- When solving inequalities with variables in the denominator, such as $\frac{1}{x} < 3$, the denominator is sometimes positive and sometimes negative. If we were to multiply both sides by x, we wouldn't know whether to reverse the inequality symbol or not. Therefore, we multiply by the **square** of the denominator.
- To solve an inequality with variables in the denominator:
  1. Multiply both sides by the square of the denominator, then solve using quadratic/cubic inequalities.
  2. Solve $\text{denominator} \neq 0$ to see if there is a restriction.

⚠ Do not cancel factors. This can cause you to change the domain of the inequality, lose critical points, or include invalid solutions.

Example Solve inequalities with variables in the denominator

Solve $\frac{1}{x} < 3$

1. $\times x^2 \quad \times x^2$

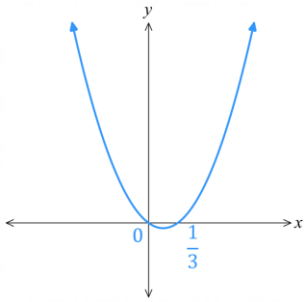
$x < 3x^2$

$0 < 3x^2 - x$

$3x^2 - x > 0$

$x(3x - 1) > 0$

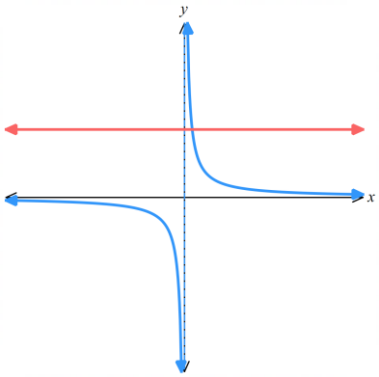
$x < 0, \quad x > \frac{1}{3}$



2. $x \neq 0$,
but this doesn't change our answer

$$\therefore x < 0, \quad x > \frac{1}{3}$$

Graphical representation:



Guided Practice

Solve:

a $\frac{3}{x} \geq 8$

b $\frac{6}{x-2} \geq 3$

c $\frac{5}{x+3} \geq 1$

d $\frac{5}{x+3} \leq 1$

e $\frac{x^2-3x+10}{x+3} \leq 2$

f $\frac{x^2-9}{x+3} \leq 1$

Do not cancel factors!

1 Solve:

a

$$\frac{1}{x} > 1$$

$0 < x < 1$

b

$$\frac{3}{x} < 1$$

$x < 0, x > 3$

c

$$4 + \frac{3}{x} \geq 0$$

$x \leq -\frac{3}{4}, x > 0$

2 Consider $\frac{2}{x+1} \leq 4$

a Write the restriction on the values of x.

$x \neq -1$

b Multiply both sides by $(x+1)^2$ to show $4(x+1)^2 - 2(x+1) \geq 0$

c Factorise out $(x+1)$ and show this simplifies to $(x+1)(2x+1) \geq 0$

d By solving the quadratic inequality, solve $\frac{2}{x+1} \leq 4$

(remember to check the restrictions)

$x < -1, x \geq -\frac{1}{2}$

3 Solve

a

$$\frac{2}{x+1} \leq 1$$

$x < -1, x \geq 1$

b

$$\frac{2}{x-3} > 1$$

$3 < x < 5$

c

$$\frac{3}{x+4} \geq 2$$

$-4 < x \leq -\frac{5}{2}$

d

$$\frac{5}{2x-3} < 1$$

$x < \frac{3}{2}, x > 4$

e

$$\frac{2}{3-x} > 1$$

$1 < x < 3$

f

$$\frac{4}{5-3x} \leq -1$$

$\frac{5}{3} < x \leq 3$

4 Solve

a

$$\frac{5x}{2x-1} \geq 3$$

$\frac{1}{2} < x \leq 3$

b

$$\frac{2x-3}{x+4} \leq 1$$

$-4 < x \leq 7$

c

$$\frac{2x+5}{x+3} < 1$$

$-3 < x < -2$

d

$$\frac{x(x+3)}{x+1} < 2$$

$x < -2, -1 < x < 1$

e

$$\frac{x^2-4}{x-2} > 0$$

$x > -2, x \neq 2$

f

$$\frac{4}{x-3} \leq x$$

$-1 \leq x < 3, x \geq 4$

g

$$\frac{1}{x} \leq \frac{1}{x+2}$$

$-2 < x < 0$

h

$$x + \frac{1}{x} \geq 2$$

$x > 0$

5 Solve this polynomial inequality by sketching the graph.

$$x(x+2)(x-1)(x+4) \geq 0$$

$x \leq -4, \quad -2 \leq x \leq 0, \quad x \geq 1$

6 Solve $1 + 2x - x^2 \geq \frac{2}{x}$

$x^2 + 2x^3 - x^4 \geq 2x$

$-x^4 + 2x^3 + x^2 - 2x \geq 0$

$x^4 - 2x^3 - x^2 + 2x \leq 0$

$x^3(x-2) - x(x-2) \leq 0$

$(x-2)(x^3 - x) \leq 0$

$x(x-2)(x^2 - 1) \leq 0$

$x(x-2)(x-1)(x+1) \leq 0$

By sketching the polynomial, $-1 < x < 0, \quad 1 < x < 2$

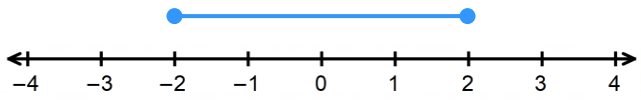
Absolute Value Inequalities

Number Line Representation of Absolute Value Inequalities

$|x| \leq 2$ means:

'the distance of x from 0 is 2 or less'

Use the number line to answer the question:



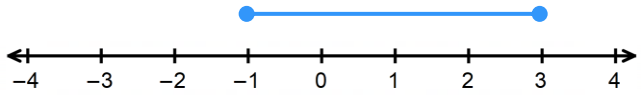
$|x| \leq 2$,

$2 \leq x \leq 2$

$|x - 1| \leq 2$ means:

'the distance of x from 1 is 2 or less'

Use the number line to answer the question:



$|x - 1| \leq 2$,

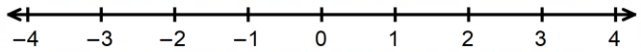
…… $\leq x \leq$ ……

$|2x - 3| \leq 2$

This can be rewritten as $|x -$ …… $| \geq$ ……

'the distance of x from …… is ……..……'

Use the number line to answer the question:

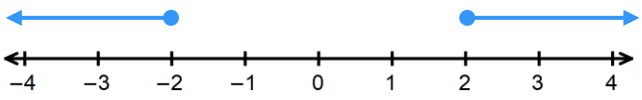


$|2x - 3| \leq 2$,

$|x| > 2$ means:

'the distance of x from zero is ……'

Use the number line to answer the question:

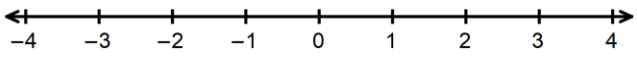


$|x| > 2$,

$|x - 1| \geq 2$ means:

'……';

Use the number line to answer the question:



$|x - 1| \geq 2$,

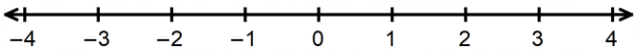
……

$|2x - 3| \geq 2$

This can be rewritten as $|x -$ …… $| \geq$ ……

'the distance of x from …… is ……..……'

Use the number line to answer the question:



$|2x - 3| \geq 2$,

Complete the sentence:

If the question is $|f(x)| < a$, then the answer is ……

If the question is $|f(x)| > a$, then the answer is ……

Solving Absolute Value Inequalities

When $|f(x)| \leq a$ (or $\geq a$), split into two cases: $f(x) \leq a$ or $-f(x) \leq a$ and solve.

When $|f(x)| \leq g(x)$, you can solve it graphically, square both sides, or consider cases.

⚠ Do **not** do $f(x) \leq \pm a$ as this means the inequality direction will be incorrect.
⚠ Generally, you are not required to square both sides or consider cases in the HSC course.

Example Solve absolute value inequalities

Solve

$$|2x - 4| < 10$$

$2x - 4 < 10$ | $-(2x - 4) < 10$

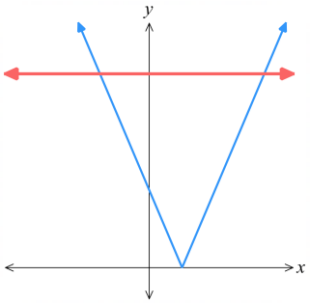
$2x < 14$ | $-2x + 4 < 10$

$x < 7$ | $-2x < 6$

$x > 3$

$\therefore 3 < x < 7$

Graphical representation:



Guided Practice

Solve, and sketch the graphical representation.

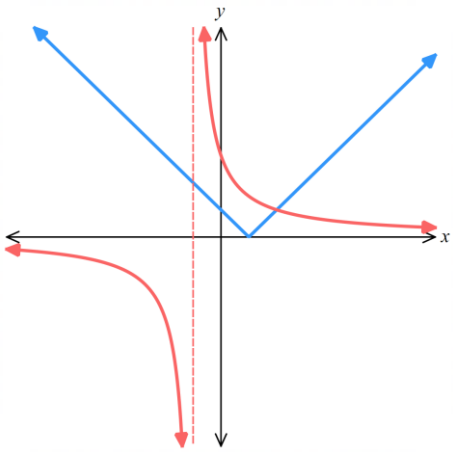
a $|x - 5| \leq 1$

b $|5x + 7| \geq 3$

c $|4x - 5| > 2x + 3$

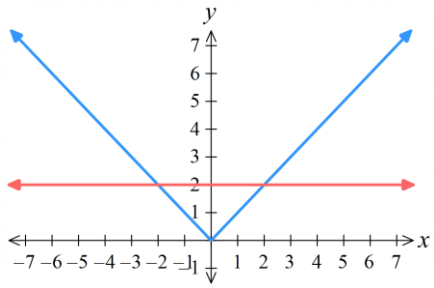
d $|x - 1| \geq \frac{3}{x+1}$

Solve this graphically by considering the points of intersection.



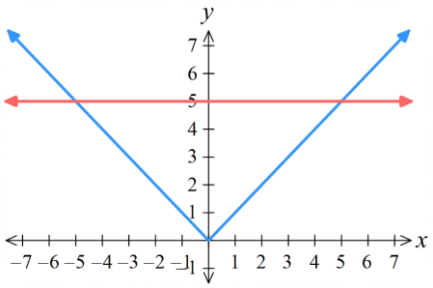
1 Solve these inequalities. A graphical representation is given.

a $|x| \leq 2$



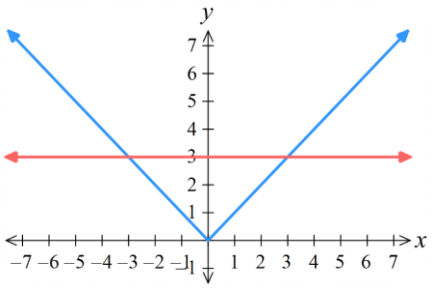
$-2 \leq x \leq 2$

b $|x| > 5$



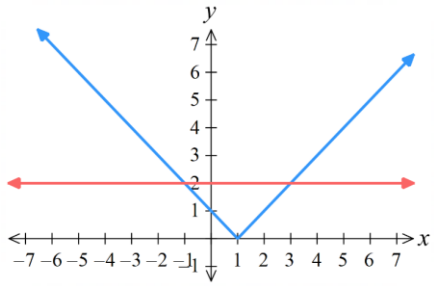
$x < -5, x > 5$

c $|x| < 3$



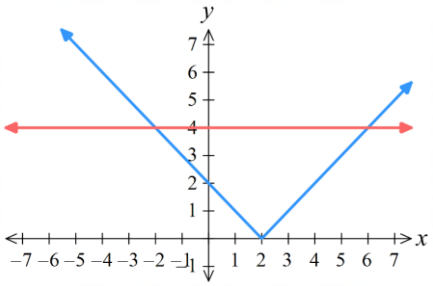
$-3 < x < 3$

d $|x - 1| < 2$



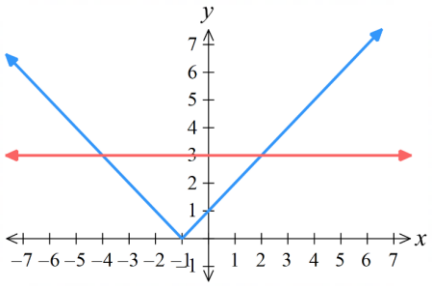
$-1 < x < 3$

e $|x - 2| \geq 4$



$x \leq -2, \ x \geq 6$

f $|x + 1| > 3$



$x < -4, \ x > 2$

2 Solve these inequalities algebraically.

a $|x| \geq 4$

$x \leq -4, \ x \geq 4$

b $|x + 1| < 3$

$-4 < x < 2$

c $|x - 3| \geq 2$

$x \geq 5, \ x \leq 1$

d $|x - 2| < 3$

$-1 < x < 5$

e $|3x - 5| \leq 4$

$\frac{1}{3} \leq x \leq 3$

f $|x - 7| \geq 2$

$x \geq 9, x \leq 5$

g $|2x + 1| < 3$

$-2 < x < 1$

h $|6x - 7| > 5$

$x < \frac{1}{3}, x > 2$

i $|5x + 4| \geq 6$

$x \geq \frac{2}{5}, \ x \leq -2$

3 Solve:

a $|6x - 7| > 5$

$x > 2,\ x < \frac{1}{3}$

b $|7 - 6x| > 5$

$x > 2,\ x < \frac{1}{3}$

Explain why the answers are the same.

Due to the identity $|ax - b| = |b - ax|$

4 Solve:

a $\left|\frac{a}{2} - 3\right| \leq 2$

$2 \leq a \leq 10$

b $14 > |8 - 2x|$

$-3 < x < 11$

c $11 \leq |6 - 5n|$

$n \leq -1,\ n \geq \frac{17}{5}$

5 Explain why:

a $|ax + b| \geq 0$ is true for all real values of x.

Absolute value cannot be negative,
so this inequality is true for all real x

b $|ax + b| > 0$ is true for all real values of x, except $x = -\frac{b}{a}$

$ax + b \neq 0$

$\therefore x \neq -\frac{b}{a}$

c $|ax + b| < 0$ has no solution,

Absolute value cannot be negative,
so this inequality is false for all real x

d $|ax + b| \leq 0$ has only one solution.

Absolute value cannot be negative.
so this inequality is only true when $ax + b = 0; x = -\frac{b}{a}$

6 Solve these inequalities.

a $\lvert x+5\rvert \geq 0$ All real x	**b** $\lvert 4-x\rvert \leq 0$ $x=4$	**c** $\lvert 3x-4\rvert > 0$ All real x except $x \neq \frac{4}{3}$
d $\lvert 4-3x\rvert < 0$ No solution	**e** $\lvert 4-x\rvert \leq 0$ $x=4$	**f** $\lvert 3x+2\rvert < -7$ No solution
g $\lvert 3x-4\rvert > -1$ All real x	**h** $\lvert 2x+5\rvert < -2$ No solution	**i** $\lvert 3x+5\rvert \leq 0$ $x=\frac{5}{3}$

7 Solve

$$\frac{1}{\lvert 2x-1\rvert} < 2$$

Reciprocate both sides

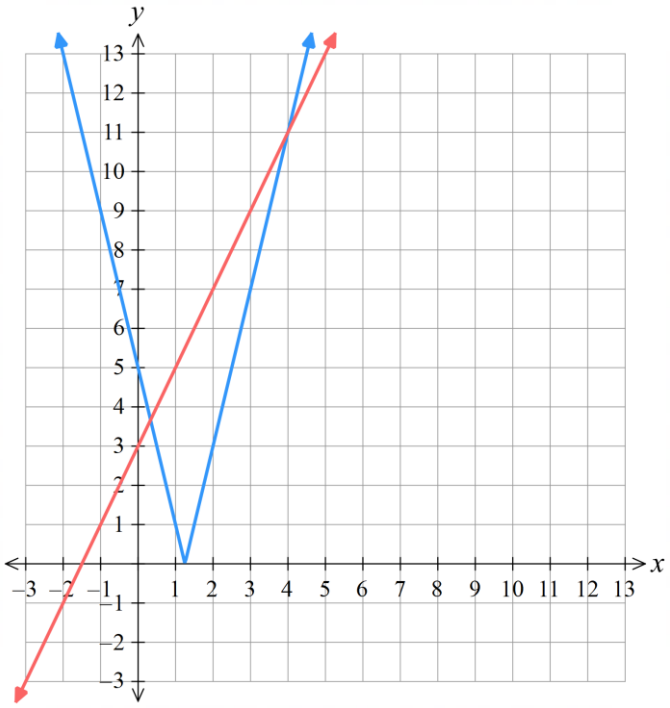
$\lvert 2x-1\rvert > \frac{1}{2}$

$2x-1 > \frac{1}{2}, \quad -2x+1 > \frac{1}{2}$

$x > \frac{3}{4}, \quad x < \frac{1}{4}$

The next questions involve solving an absolute value inequality where there is a function on both sides. We must either solve it graphically or consider cases.

8 Sketch $y = \lvert 4x-5\rvert$ and $y = 2x+3$ on the same axis, hence solve $\lvert 4x-5\rvert > 2x+3$.

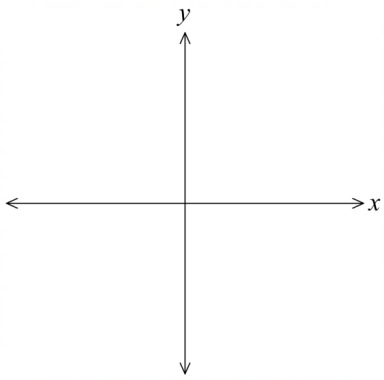


By observing when the graph of $\lvert 4x-5\rvert$ is above $2x+3$

$x > 4, \quad x < \frac{1}{3}$

9 Solve $|2x + 1| \leq |x - 4|$ by following these steps:

a Sketch both curves, try to be as accurate as possible.



b Identify which "branches" of the graphs intersect.

$y = 2x + 1$ intersects with $y = -(x - 4)$

and $y = -(2x + 1)$ intersects with $y = -(x - 4)$

c Find the points of intersection.

$2x + 1 = -(x - 4)$

$-(2x + 1) = -(x - 4)$

The points of intersection are at $x = -5$ and 1

d Hence solve the inequality.

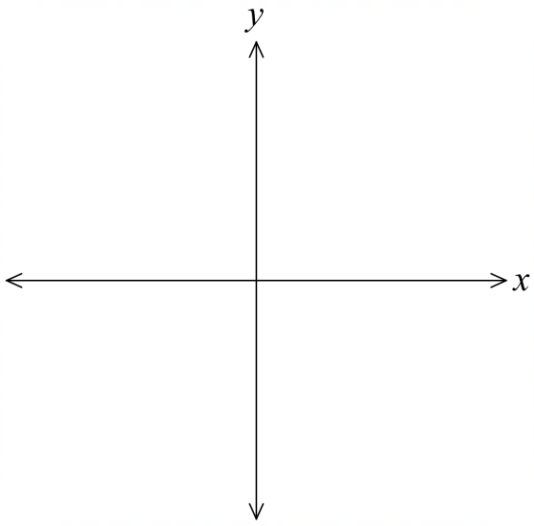
By examining the graph:

$-5 \leq x \leq 1$

10 **2008 VCE Band 4**

Solve $|2x - 1| \leq |x - 3|$.

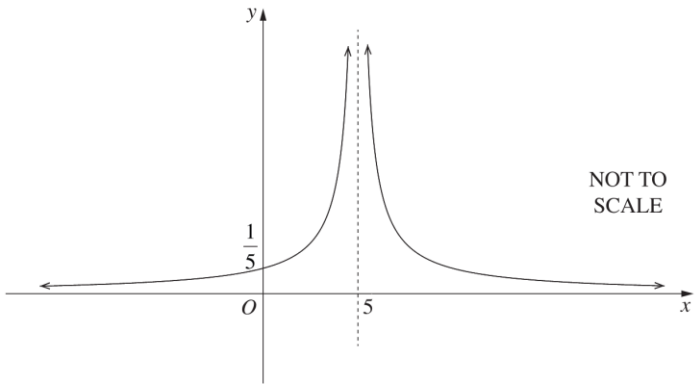
(hint: sketch the curves)



$-2 \leq x \leq \frac{4}{3}$

11 2024 HSC Extension 1 Band 4

The diagram shows the graph of $y = \dfrac{1}{|x-5|}$



Solve $\dfrac{x}{6} \geq \dfrac{1}{|x-5|}$ by following these steps:

a Show that $x|x-5| \geq 6$. Explain why we do not need to multiply by the square of $|x-5|$.

Cross multiply: $x|x-5| \geq 6$

we do not need to multiply by the square as $|x-5|$ must be positive

b Find the points of intersection by solving $x|x-5| = 6$

$$x(x-5) = 6$$
$$x^2 - 5x - 6 = 0$$
$$(x-6)(x+1) = 0$$
$$x = 6, -1$$

$$-x(x-5) = 6$$
$$-x^2 + 5x - 6 = 0$$
$$x^2 - 5x + 6 = 0$$
$$(x-2)(x-3) = 0$$
$$x = 2, 3$$

c Therefore, sketch $y = \dfrac{x}{6}$ and state the solutions to $\dfrac{x}{6} \geq \dfrac{1}{|x-5|}$

if you sketch $y = \frac{x}{6}$, there won't be an intersection point at $x = -1$

by inspecting the graph, $2 \leq x \leq 3,\ x \geq 6$

12 We will solve $\frac{x}{6} \geq \frac{1}{|x-5|}$ without a graph by considering cases.

a Show that $x|x-5| \geq 6$

Cross multiply: $x|x-5| \geq 6$
(we do not switch the inequality sign as $|x-5|$ is positive)

b Consider $|x-5|$ as the piecewise function:

$$|x-5| = \begin{cases} x-5, & \text{for } x \geq 5, \\ -(x-5), & \text{for } x < 5 \end{cases}$$

Therefore, remove the absolute value signs and solve $x|x-5| \geq 6$:
(you will need to use quadratic inequalities)

$x \geq 5$ $x < 5$:

For $x \geq 5$: $x(x-5) \geq 6$
$x^2 - 5x - 6 \geq 0$
$(x-6)(x+1) \geq 0$
$x \geq 6, \ x \leq -1$
But $x \leq -1$ is not a solution as $x \geq 5$

For $x < 5$: $-x(x-5) \geq 6$
$-x^2 + 5x - 6 \geq 0$
$x^2 - 5x + 6 \leq 0$
$(x-2)(x-3) \leq 0$
$2 \leq x \leq 3$

c Explain why $x \leq -1$ is not a solution to the inequality.

We found $x \leq -1$ when we considered the case $x \geq 5$, which is a contradiction.

d Therefore, state the solution to the inequality $\frac{x}{6} \geq \frac{1}{|x-5|}$

$\therefore 2 \leq x \leq 3, \quad x \geq 6$

13 **2020 VCE Band 4**

Solve $3 - x > \frac{1}{|x-4|}$

$(3-x)|x-4| > 1$

For $x > 4$, $(3-x)(x-4) > 1$

$-x^2 + 7x - 12 > 1$

$-x^2 + 7x - 13 > 0$

No solution as $\Delta < 0$

For $x < 4$, $-(3-x)(x-4) > 1$

$x^2 - 7x + 11 > 0$

$x < \frac{7-\sqrt{5}}{2}, x > \frac{7+\sqrt{5}}{2}$

But since this is for the case $x < 4$,

$x < \frac{7-\sqrt{5}}{2}$ is the only solution.

14 Consider the inequality

$$\left|x+\frac{1}{x}\right|<2x$$

a Explain why x must be positive.

Since $\left|x+\frac{1}{x}\right|>0,\ 2x>0,$ so $x>0$

b Hence explain why $x+\frac{1}{x}$ must be positive.

A positive number plus its reciprocal must also be positive.

c Solve the inequality.

$x+\frac{1}{x}<2x$

$x^3+x<2x^3$

$-x^3+x<0$

$x^3-x>0$

$x(x^2-1)>0$

$x(x-1)(x+1)>0$

$-1<x<0,\ x>1$

But since $x>0, x>1$ is the only correct solution.

since $x+\frac{1}{x}>0$, we do not need to consider the case where $x+\frac{1}{x}<0$

$\therefore x>1$ is the only solution.

15 Solve $|x|-2<4-x^2$

$|x|<6-x^2$

$x>0: x<6-x^2$

$x^2+x-6<0$

$(x+3)(x-2)<0$

$-3<x<2$

$x<0: -x<6-x^2$

$x^2-x+6<0$

$(x-3)(x+2)<0$

$-2<x<3$

The inequality that satisfies both $-3<x<2$ and $-2<x<3$ is

$-2<x<2$